Solutions and teaching notes for chapter 3

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $(3;-4)$. *Action:* Subtract end minus start on two drawings.
2. $\sqrt{3^2+(-4)^2}=5$. *Action:* Review Pythagoras and distinguish norm from coordinates.
3. $\overrightarrow{AC}$. *Action:* Draw the composite displacement.
4. $a^2+2ab+b^2$. *Action:* Identify the cross term.
5. $\dfrac{3}{5}$. *Action:* Name the sides relative to the angle.
6. $(1;2)$. *Action:* Average the two coordinates separately.

Investigation activity

Take points on the circle with diameter AB; proof via the midpoint I: product MI^2-9. The endpoints are included because one of the vectors is zero.

Immediate applications

1. 0 then -10. The cosine is 0 and -1 respectively.
2. $3t-12=0$, so $t=4$.
3. The square is $4-2+9=11$; the norm is $\sqrt{11}$.
4. $32=16+16-32\cos A$, so $\cos A=0$ and $A=90°$.
5. $MI^2-9=7$: circle with centre $(3;0)$ and radius 4.

Exercises: method and results

Exercise 3.1 – Direct calculations. Product $4-6=-2$; norms $\sqrt{5}$ and 5; sum $(5;1)$ with square 26. Check: $5+2(-2)+25=26$.

Exercise 3.2 – A right triangle? $\overrightarrow{AB}=(3;1)$, $\overrightarrow{AC}=(-1;3)$; product 0. $AB=AC=\sqrt{10}$, area $\dfrac{AB\times AC}{2}=5$ area units.

Exercise 3.3 – Length without a frame. $\overrightarrow{AC}=\overrightarrow{AB}+\overrightarrow{AD}$ and $\overrightarrow{BD}=\overrightarrow{AD}-\overrightarrow{AB}$. Product $\dfrac{4\times 3}{2}=6$; $AC^2=16+9+12=37$ and $BD^2=16+9-12=13$.

Exercise 3.4 – Recovering an angle. $\cos\theta=-\dfrac{6}{12}=-\dfrac{1}{2}$; in $[0;180°]$, the angle is $120°$. The domain ensures uniqueness.

Exercise 3.5 – A variable locus. $MI^2 = k + \dfrac{AB^2}{4}$. If the right-hand side is negative: empty set; zero: the single point I; positive: circle whose radius is its square root. The discussion bears on a squared distance.

Exercise 3.6 – Work done by a force. $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. At 120°, $W = -100$ J. The sign distinguishes help to the displacement from opposition; it does not describe a negative distance.

Exercise 3.7 – Contrapositive. The direct property "orthogonal implies equal squares" follows from the expansion. Its contrapositive is exactly the statement. We can also write $2\vec{u} \cdot \vec{v} \neq 0$.

Exercise 3.8 – Delayed review. Al-Kashi: $\dfrac{9 + 16 - 25}{2} = 0$. Converse of Pythagoras: triangle right-angled at A, so zero product. Name the hypotheses of each method.

CQFD, exercise 9 p. 250, questions a and b. a: $1 \times 3 + 2 \times (-1) = 1$. b: $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. $\overrightarrow{AB} = (4;6)$, $\overrightarrow{AC} = (-3;4)$, $I(-1;-2)$ and $\overrightarrow{CI} = (5;-1)$. Hence $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI} = 14$. Chasles gives $\overrightarrow{AB} \cdot \overrightarrow{AI} = 12 + 14 = 26$. Two checks: $\overrightarrow{AI} = (2;3)$ gives $8 + 18 = 26$; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16 + 36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.
2. C. Square root of 25.
3. D. Scalar criterion.
4. B. $\dfrac{2 \times 3}{2} = 3$.
5. C. Expand the scalar square.

MCQ 2.

1. D. Negative cosine.
2. A. Coordinates by projection.
3. C. $MI^2 = \dfrac{AB^2}{4}$.
4. B. Bilinearity: $2(-1)3$.
5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.
2. True: its product with every vector is 0.
3. False: $(1;0)$ and $(0;1)$ are nonzero and orthogonal.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.